UNIT 15 BASIC CONCEPTS OF METRIC SPACE AND POINT SET TOPOLOGY

Structure

15.0 OBJECTIVES

After going through this unit, you will be able to:

- define 'metric space', and explain how it generalises the concept of Euclidean space;
- define 'topological space', and explain how it generalises each of the concepts of metric space and Euclidean space;
- explain the significance and applications of each of the disciplines of 'metric space' and 'topology',

- define the concepts of open interval, closed interval, open set and closed set in a metric space, and tell about their properties.
- define the concepts of convergence of a sequence, continuity of a function, and connectedness and compactness of a set in a metric space, and use these in solving relevant problems
- define the concept of 'topology', or, 'topological space'
- define the concepts of continuity of a function, connectedness, and compactness of a set in a topological space, and use these in solving relevant problems

15.1 INTRODUCTION

Generalization is the hallmark of Modern Mathematics. Since the beginning of 20^{th} century, a significant part of the developments in Mathematics have taken place through generalizations of earlier concepts and disciplines in Mathematics. Metric space and topology are disciplines which are generalizations of some of the disciplines studied earlier: (i) *'metric space'* being *generalisation **to** arbitrary sets **of** Euclidean spaces* (R^2, and, in general R^n) studied in the previous unit; and (ii) *'topology'* being generalization of geometry, and even of metric space.

The concept of ***'Metric'*** is *formalisation (i. e. expression in mathematical terms and symbols)* of the *intuitive* concept of *distance* in our daily lives. The concept of ***distance*** occurs

- not only spatially, say as, distance between two cities, or as between two stars; ***it also occurs***
- temporally as difference of number of days in between, say, of two dates of a year;
- as difference in the average temperatures, say, in May and December in a particular place;
- in human relations, say,
    - as from sibling *(at distance 1)* to first cousin *(at distance 2)* to second cousin *(at distance 3)* etc.; or
    - as generational difference from parents *(at distance 1)* to grand-parents *(at distance 2)* to great-grand-parents *(at distance 3)* etc.

The parenthetical phrases above give an idea of formalization/ metrification of the corresponding distance concepts.

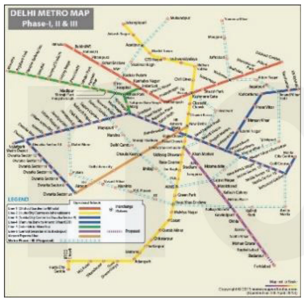
The concept of distance and its mathematical version 'metric' have now become an essential tool in quantitative analysis for almost every academic discipline and its applications including probability, statistics, graph theory, clustering, data analysis, pattern recognition, computer graphics, image analysis, speech recognition, and information retrieval, astronomy, molecular biology etc.

For example, Genetic distance measures genetic divergence between species or between populations within a species.

In economics, the ***economic distance*** (Patel, 1965) between two countries is the time (in years) for a lagging country to catch up to the same per capita income level as the present one of an advanced country. Some of the well-known ***Production Economics distances*** include the ***directional distance function,*** the ***Shephard input distance function,*** and the ***distance to frontier.***

Those who are interested in the finer applications of the concept of *distance* may refer to the tome by Deza and Deza (2016).

Some introductory remarks on topology: Look at the two figures below, which may serve as nice introduction to topology and its applications. The LHS gives a sort of aerial view of Delhi Metro Network, and the RHS shows a type of map available within a Metro rail.



The differences are (i) the curved lines, and angles between them, of LHS are replaced on RHS by straight lines with almost uniform angles between consecutive pairs of them, and (ii) any two consecutive halts/ stations—despite possibly having quite different distances between them—are shown equidistant in the map on RHS.

By losing the sense of exact distance and exact angle, we get a map which is much more comprehensible to the travellers. *(The above idea of maps is borrowed from Richard Earl)*

These are some of the ideas which lead us to the discipline of topology from Euclidean geometry.

Topology is also known as Rubber-Sheet Geometry, by which, in the figure on RHS, the cup on the left and the doughnut on the right, obtained from cup, through continuous transformations, are taken topologically equivalent, in the same sense as two congruent triangles are taken equivalent in Euclidean geometry.

For the similar reasons, the four letters **E, F, T, Y** are taken *topologically equivalent* in the sense that each can be obtained from the other by stretching/ shrinking the rubber on which any one of these is supposed to be written.

Topology has applications to a number areas in economics, including the study of *General Equilibrium,* a branch of economic theory that studies the equilibrium state of an economy inhabited by agents with different, and sometimes conflicting, interests. *Nash equilibrium* is well-known. For more details, in this respect, one may refer to the beautiful article *Topology and Economics* by *Songzi Du.*

The word *topology* is derived from the Greek words '*topo*' meaning *'place, or location*', and '*logy*', meaning '*study*'. It is concerned with the properties of a geometric object that are preserved under continuous deformations, such as stretching, twisting, crumpling, and bending.

The working definition and the rest of the discussion on topology follow in section 15.3.

Before we discuss the topics in finer details, let us know the origins of the two, quite closely related, disciplines:

Maurice Fréchet, a French mathematician, introduced in 1906, the notion of '***Metric Space***' through his outstanding thesis *Sur Quelques Points du Calcul Fonctionnel* (within a systematic study of functional operations). And, ***Felix Hausdorff***, a German mathematician, published in 1914, his famous *Grundzüge der Mengenlehre* (Foundations of Set Theory) where the theory of ***topological and metric spaces*** (*metrische Räume*) was created.

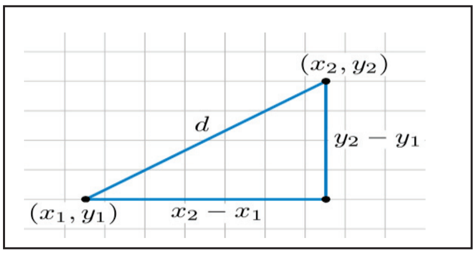
In this unit, first we generalize some of the important concepts, e. g. convergence and continuity of Euclidean space to a metric space, and then to topological space. We initiate our discussion with the topic of Metric Space.

15.2 METRIC SPACE

In practical applications, we may need more than one metric or formal distance between the same pair of points.

15.2.1 Metrics, the Formalised Distances; and some well-known Metrics

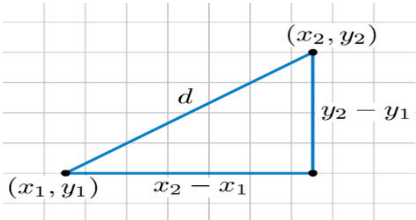
We are already familiar with the ***Euclidean distance/ metric*** in R^2 (or, even in R^n), by which distance $\mathbf{d_E}$ between two points P (x_1, y_1) and Q (x_2, y_2), as shown in the figure below on LHS, is given by the formula on RHS

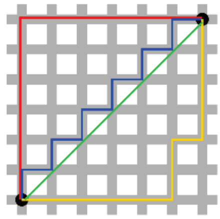


$$\mathbf{d_E\ (P, Q) = \sqrt{[(x1 - x_2)^2 + (y_1 - y_2)^2]}}$$

However, this is not the (minimum) distance actually covered by a *road taxi* in a city. Theoretically, we assume the roads in a city are mutually parallel and perpendicular, somewhat like in the figure on RHS below.

Then the (minimum) distance, which is called ***Taxicab distance,*** $\mathbf{d_T}$, is the sum of the distance covered along x-axis and the distance covered along y-axis, as shown below in the figure on LHS, and in *red colour* in figure on RHS,





Symbolically,

Taxicab distance, d_T between points $x= (x_1, y_1)$ and $y= (x_2, y_2)$ = $d_T (x, y) = |x_1 - x_2| + |y_1 - y_2|$.

The Taxicab distance is also called ***Manhattan distance.*** Both Manhattan distance and Euclidean distance are special cases of ***Minkowski distance***. In general, *Minkowski distance of order p* for R^n, the n-dimensional space over real

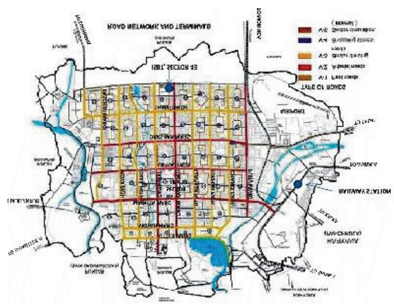
$$\left(\sum_{i=1}^{n} |x_i - y_i|^p\right)^{1/p}$$ for order p= 1, 2, 3, …

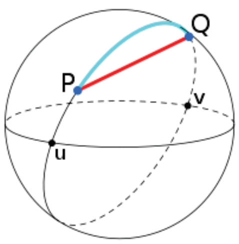
numbers, is given by

***Manhattan/Taxicab** distance is the Minkowski distance of **order one**, and **Euclidean distance** is the Minkowski distance of **order 2**.*

Minkowski distances of different orders, including Euclidean distance and Manhattan distance, find applications in a number of domains including statistics and for optimization problems in various domains including economics. In practical applications, some distances are more appropriate than others depending on application domain.

For example, in the matter of urban planning, Taxicab distance is more useful than Euclidean distance: planning of the medical, educational and other facilities in a city, depends more on the mass-transit facilities, including road and metro network, which involve Taxicab distance, rather than the Euclidean distance. *(For example, Road Map of Chandigarh on RHS)*





Similarly, the distance between two points, say P and Q, on the surface of earth may be considered alternatively as ***(i)*** the length of the shortest path *along (the great circle of) the surface*, (in blue colour) which is used for shipping and aviation, and as ***(ii)*** the straight-line distance through the Earth's interior (in red colour). The latter notion is used, for example, in seismology for measuring the length of time taken for seismic waves to travel (Wikipedia).

Summarising the above discussion, for the same set of points, there may be many metrics giving different values depending upon the problem domain. However, *there cannot be any* ***appropriate measure*** which gives *distance between Delhi and Chennai by direct train* ***to be more than*** *that of distance of Delhi and Chennai via the moon*. So, any such measure *cannot* be a metric. ***Hence, the question arises that what should be the criteria for a relation/ function/measure to be called a metric on an arbitrary set.***

Mathematics, which involves both the art of abstraction, and the science of reasoning in the world of abstracts, decides—taking into consideration the above discussion—the set of desired criteria, and, which in turn, lead to a very useful mathematical discipline, viz. ***Metric Space***, discussed in the next subsection.

15.2.2 Metric Space

We start with the fundamental concept of Metric space, viz.

15.2.2.1 Metric on a Set: For a given non-empty set, say S, and a ***function***, (generally, denoted by d) d: S x S → R, the set of real numbers, *which, for x, and y ∈ S, satisfies the following conditions, is called a metric*

i) d (x, y) ≥ 0, *(the cost of going from any point to any point is non-negative)*

ii) d (x, y) = 0 if and only if x =y, i. e.

  a. d (x, x) = 0, for any x in S, *(it does not cost to remain at the same point)* and

  b. d (x, y) > 0, if x ≠ y, *(the cost of going to a different point is positive)*,

iii) d (x, y) = d (y, x) *(symmetry property: the cost of going from x to y is the same as the cost of going back to x from y)*

iv) d (x, z) ≤ d (x, y) + d (y, z) *(transitive property: the cost of going directly from x to z does not exceed the cost of going from x through y to z)*

15.2.2.2 Metric Space: The ordered pair (S, d), where set S $\neq \emptyset$, and d is a metric on S, is called a metric space.

Example 15.1: In Unit 3, we introduced the mod function as follows

Mod: R→ R such that

$$\text{Mod}(x) = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$$

Thus, Mod (x) =|x| ≥ 0

The above function can be used to define a metric d on R, as follows

d: R x R → R, such that

d (x, y) = |x − y|

We show (R, d) is a metric space, as d satisfies the following properties:

i) As x and y ∈ R, then x–y ∈ R, hence, by definition of Mod, d (x, y) = |x − y| ≥ **0**

ii) Also, we know, **|x| = 0, if and only if x** = 0.

Therefore,

d (x, y) = |x − y| =0 if and only if (x–y) =0, i. e. iff x =y

iii) **(symmetry)**

d (x, y) = |x − y| = |y − x| (by definition of mod function)

= **d (y, x)** (by definition of d)

iv) **(Triangular Inequality)**

For x, y, z ∈ R, consider

d (x, z) = |x − z| = | (x− y) + (y − z)|

≤ | (x− y)| + | (y − z)| (by triangular property of mod)

= d (x, y) + d (y, z) = (by definition of d).

Hence, the proof.

Example 15.2: Let S be any non-empty set, define function

d: S x S → R, such that for x, y ∈ S,

$$\mathbf{d(x, y)} = \begin{cases} \mathbf{0, \text{ if } x = y} \\ \mathbf{1, \text{ if } x} \neq y. \end{cases}$$

Then (S, d) is a metric space.

Solution:

i) As d (x, y) = 0 or 1, Hence, d (x, y) ≥ 0

ii) Let d (x, y) =0, then x = y (must), else d (x, y) will be 1.

iii) (symmetry) If d (x, y) = 0, then x=y, hence d (x, y) = d (x, x) = d (y, x)

else, if d (x, y) =1, then $y \neq x$, and hence, d (y, x) =1

Thus, d (x, y) = d (y, x).

iv) (Triangular inequality)

Case (a): if x =z, then

d (x, z) = 0 =0 + 0 $\leq$ d (x, y) + d (y, z)

Case (b): $x \neq z$, then d (x, z) = 1,

and hence,

for any y of S, y is not equal to at least one of x and z.

Let $x \neq y$

Then d (x, y) = 1

d (x, z) = 1 $\leq$ 1+0 $\leq$ d (x, y) + d (y, z) *(as d (y, z) is at least 0)*.

Hence the proof.

Check Your Progress 1

1) Show that (R^2, d_T) is a metric space, where, R^2 = R x R, and d_T, the Taxicab distance, is defined as follows.

For x= (x_1, x_2), and y= (y_1, y_2),

$d_T(x, y) = |x_1 - y_1| + |x_2 - y_2|$.

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2) Show that (R^2, d_E) is a metric space, where, R^2 = R x R, and d_E, the Euclidean distance, is as defined as follows.

For x= (x_1, x_2), and y= (y_1, y_2),

$$d_E(x, y) = \sqrt{[(x_1 - x_2)^2 + (y_1 - y_2)^2]}$$

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15.2.3 Open Interval/ball/sphere and Closed Interval/ ball/sphere

From Unit 1, we are familiar with the concepts of ***open and closed intervals*** in R, equipped with mod (viz. $|x|$) induced distance/metric function.

An **open interval** on real line can (always) be considered as

$]a - \varepsilon, a + \varepsilon[= \{x: | x - a | < \varepsilon\}$, for some real number a, and for $\varepsilon > 0$.

A **closed interval** on real line can (always) be considered as

$$[a - \varepsilon, a + \varepsilon] = \{x: | x - a | \leq \varepsilon\}, \text{ for some real number } a \text{ and for } \varepsilon > 0.$$

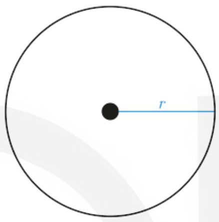
We may extend the concept of ***open interval*** to 2-dimension as an ***open ball*** as follows

Let $\mathbf{a} = (\mathbf{a_1}, \mathbf{a_2}) \in R^2$ be a point in the R^2 Euclidean space. Then***, open ball B (a, r) centred at point a and of radius r>0***, a real number**,** is the following subset of R^2

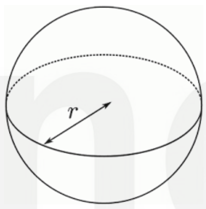
$\mathbf{B\ (a, r) = \{x = (x_1, x_2) \in R^2: d_E (x, a)} < r\}$, where d_E is the Euclidean distance on R^2. In other words, ***open ball*** B (a, r) is the set of all members of R^2 which are at a Euclidean distance of at most r from the point a.

We call it a ***ball***, as its shape in Euclidean space R^2 is of the form *as shown below on LHS*, but without boundary.

Open Ball



Open Sphere



We may further extend the concept of open interval to 3-dimension as an ***open sphere*** as follows

Let $a = (a_1, a_2, a_3) \in R^3$ be a point in the R^3 Euclidean space. Then, ***open sphere S (a, r) centred at a and of radius r>0***, a real number, is the following subset of R^3

$\mathbf{S\ (a, r) = \{x = (x_1, x_2, x_3) \in R^3: d_E (x, a)} < r\}$, where d_E is the Euclidean distance on R^3.

In other words, ***open sphere*** *S (a, R)* is the set of all members of R^3, which are at a Euclidean distance of at most r from a.

We call it a ***sphere***, as its shape is of the form *as shown above on RHS*, but without boundary.

We may further extend the concepts of interval/ ball/ sphere to the Euclidean space R^n

Let $\mathbf{a} = (\mathbf{a_1}, \ldots, \mathbf{a_n}) \in R^n$ be a point in the R^n Euclidean space. Then, ***open sphere*** **S (a, r)** centred at **a** and of radius r>***0***, a real number, is the following subset of R^n

$\mathbf{S\ (a, r) = \{x = (x_1, \ldots, x_n) \in R^n: d_E (x, a)} < r\}$, where d_E is the Euclidean distance on R^n.

In other words, ***open sphere*** S(a, R) is the set of all members of R^n, which are at a Euclidean distance of at most r from the point a.

Also, the concept of ***closed interval*** may be extended to ***closed ball*** and ***closed sphere*** in respectively R^2 and R^n for $n \geq$ **3,** as follows.

B^c (a, r) = {x= (x_1, x_2) ∈R^2: d_E (x, a) ≤ r}, where d_E is the Euclidean distance on R^2.

S^c (a, r) = {x= (x_1, …, x_n) ∈R^n: d_E (x, a) ≤ r}, where d_E is the Euclidean distance on R^n.

(Please note, the notations B^c, and S^c are not standard notations for closed ball, and closed sphere)

Also, from the previous unit of this block, we know the concepts of ***open set*** and ***closed set*** *(not just open* ***interval*** *and closed* ***interval)*** as defined below, ***in R, R^2 and R^n***, each of which is equipped with Euclidean metric.

Definition Open set: A ***set*** S of real numbers is said to be ***open set, or just open***, if for each element ***a*** of S, there is a number **ε >0** such that the open interval *]a – ε, a+ ε[* ⊆ S. In other words, every member of open set S is *well inside* S, and hence, no member is a boundary point. In the previous unit, we also established

i) **Every open** *interval* **A =]a, b[is an open** *set*,

ii) **Union of open intervals is an open set.**

Definition Closed set: A set S ⊆ R is said to be a closed if

R – S = (–∞, + ∞) – S is open. R.

We may note that set of real numbers is also denoted as open interval (–∞, + ∞).

Next, we generalize these concepts to a metric space, i. e. to the system (X, d), where X is an arbitrary non-empty set, and d is a metric on X.

15.2.3.1 Open sphere, Closed sphere, Open set, Closed set of Metric space

15.2.3.1.1 Open sphere, Closed sphere of arbitrary metric space

Let (X, d) be a metric space, where set X ≠ Ø, and d is a metric on X. Then

a) ***The open sphere with centre a and radius r>0***, a real number, ***is the set***

$$S (a, r) = \{x \in X: d (a, x) < r\}.$$

b) ***The closed sphere with centre a and radius r >0***, a real number, ***is the set***

$$S^c (a, r) = \{x \in X: d (a, x) \leq r\}$$

(Please note (i) the term 'ball' is also used instead of 'sphere' (ii) the notations S^c is not standard notation for closed ball, and closed sphere)

Our examples are based on examples of previous subsection (viz.15.2.2.2) and Check Your Progress 1.

Example 15.3: For the metric space (R, d), with R, the set of real numbers, and

d: R x R → R, such that